Daily Drilling Report
SPD19B-03
3/14/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
4.0	24"	10.00	72.83	31.70	96.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
26" CP, 178.00mKB		12,632.40	606,913.73		2.60
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
183.00	357.00	355.46	174.00	12.55	13.9

Operations at Report Time
Drilling @ 420m.

Operations Summary
RIH Kick off BHA#2 f/ 27m to 178m. Tag and D/O 26" shoe and clean out rat hole to 183m. Cont' drilling to 200m. Start kick off at 200m and cont' directional drilling 24" hole to 357m.

Operations next 24hrs
Cont' drilling 24" hole to loss zone.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammad Ali Rezaei	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Hamed Daneshkhah	Drilling Engineer
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	3/31/2014
BOP Test		
Chemical Spill Drill		
Fire Drill	3/9/2014	3/11/2014
H2S Drill		
Heli Crash Drill	2/20/2014	3/14/2014
Medical Drill		
MOB Drill	3/1/2014	3/23/2014
Pit Drill		
Trip Drill		

STOP Cards

Number of Reports: 42

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:30	0.50	24"	D	12	P	Connect TDS and perform Motor & MWD shallow test w/ 600 gpm/ 1000 psi, Ok.
00:30	03:30	3.00	24"	D	04	P	P/U and M/U rest of BHA consists of: 1 jt. 9 1/2" NMDC + 1 Std 9 1/2" DC + X/O + 5 Jts 8 1/4" DC + 8" COR drilling jar (SN: COR2986AD) + 2 Jts 8 1/4" DC + X/O and RIH on 5 7/8" HWDP to 178m.Tag 26" Shoe @ 178 m.
03:30	05:30	2.00	24"	D	03	P	D/O 26" CP shoe @ 178m w/ 5 - 15 Kips/ 10 rpm/ 600 - 800 gpm/ 1000-1950 psi/ 2 - 8 Kft-lbs and clean out CMT inside rat hole to 183m. *** Reciprocate string with & without rotation inside shoe. *** Pump 50 bbls GG pill and circulate out while drilling @ 180m.
05:30	07:00	1.50	24"	D	03	P	Drill 24" hole section f/ 183m to 200m w/ 5 - 10 Kips/ 10 rpm/ 850 gpm/ 2150 psi/ 2 - 6 Kft-lbs. Spot 50 bbls PHG pill prior to run Gyro.
07:00	08:00	1.00	24"	D	11	P	Run Gyro tool to 164.06m (UBHO depth) and take survey shot. Orient TF as per DD instruction. POOH Gyro to surface. Depth (m) Inc. Azimuth (°) Gyro TF (°) 164.06 186.25 188.68 (Before orient tool face) 164.06 189.15 230.93 (After orient tool face)
08:00	09:00	1.00	24"	D	24	N	(NPT for Rig) Found problem in tool joint breaker and saver sub of TDS, work on it and rectify same.
09:00	10:30	1.50	24"	D	03	P	Directional drilling 24" hole section in slide mode f/ 200m to 221m w/ 600 gpm / 1000 psi / 10 kips. Off-line job: Top Job#2 on SP19B-04: L/U CMT unit to 26" x 18 5/8" annulus and pump 8 bbls of 12.5 ppg CMTslurry, observe annulus again fill up.
10:30	11:30	1.00	24"	D	11	P	Due to interference in MWD reading, Run Gyro tool to 190.86m (UBHO depth) and take survey shot. Depth (m) Inc.(°) Azimuth (°) Gyro TF (°) 190.86 0.7 210.28 116

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PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
11:30	00:00	12....	24"	D	03	P	Cont' directional drilling 24" hole section in slide & rotation mode f/ 221m to 357m w/ 600-900 gpm / 1000-2350 psi / 10-15 kips/ 10 rpm & 3 - 7 Kft-lbs. Slide Intervals Rotation Intervals 221m - 241m 241m - 250m 250m - 272m 272m - 285m 285m - 301m 301m - 319m 319m - 326m 326m - 347m 347m - 354m 354m - 357m >>> Sweeping Sequence: ** Pump 20 bbl GG at middle of each jt. ** Pump 20 bbl PHG at the end of each jt. ** Pump 20 bbl GG + 20 bbl PHG at the end of stand and circulate out. ** Take Survey before connection. ** Spot 30 bbl PHG pill before connection.
00:00	00:00		24"				Mar 15, 2014 00:00 - 06:00 Cont' directional drilling 24" hole section in slide & rotation mode f/ 357m to 420m w/ 800-950 gpm / 1750-2450 psi / 15-17 kips/ 10 rpm & 2 - 8 Kft-lbs. Rotation Intervals Slide Intervals 357m - 376m 376m - 390m 390m - 420m

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
357.00	Sea Water	8.60				
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.50	7.	206	3.0	1,007.0	23.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
F-GVEF	Chopper	3/14/2014 13:30	3/14/2014 13:35	Pax in: 0 & Pax out: 5
Morshed	Supply Boat	3/15/2014 02:10		Arrived DD6 location from Aban8 at 0210hrs,transferring FO to DP3 at anchorage.
Borkan	Supply Boat	3/14/2014 23:15		Delivered cargo. Rig CMT hose chocked after transferred 2 MT. Proceeded to DP3 to transfer Water-150
Navis King	Supply Boat	3/14/2014 15:15		Delivered Cargo, Barite-40MT & DW-100MT . Dep to Sahar2.
Yaghoub	Supply Boat	3/11/2014 00:50		Delivered CMT-100 MT & DW-50 MT, B/L for Kish & Departed for Sahar-2 at 1345 Hrs.
DP3	Supply Boat	2/23/2014 23:30		Delivered 100 MT DW, Received DD6 Storage cargo, Standby Anchorage.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Bentonite Bulk	15.0	105.0	90.0
Bio-Cide	5.0	139.0	99.0
Caustic Soda	6.0	427.0	267.0
Lime	4.0	150.0	68.0
Citric Acid	8.0	210.0	90.0
Guar Gam MP	60.0	1,421.0	1,202.0
Soda Ash	6.0	151.0	54.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Potable Water	22.0	17.0	79.0
Cement G	2.0	102.0	155.0
Diesel Fuel	9.0	0.0	162.0
Drill Water	100.0	250.0	550.0
Bentonite Bulk	9.0	0.0	82.0
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	40.0	209.0
Bentonite BB	0.0	0.0	45.0

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Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
2	24	Varel	ETR1GJMRS	1239954	Mill Tooth	115	20/20/20/20
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.22	12.55	70,710.00	183.00	357.00	174.00	13.9	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
600	900	1,000.0	2,300.0	10	15	69	114
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
16.86	5.61	44.70	55.30	217	215	195	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
183.00	200.00	0.83	3,000.0	900	10	2,300.0	10	197
200.00	241.00	3.80		600		1,000.0	10	197
241.00	250.00	0.43	3,000.0	800	10	1,850.0	10	204
250.00	272.00	1.24	3,000.0	900		2,300.0	10	204
272.00	285.00	0.77	3,000.0	800	10	1,850.0	12	204
285.00	301.00	0.76		900		2,300.0	10	204
301.00	319.00	1.27	3,000.0	900	10	2,300.0	10	210
319.00	326.00	0.50		900		2,300.0	12	210
326.00	347.00	1.92	3,000.0	800	10	1,850.0	12	215
347.00	354.00	0.64		900		2,300.0	14	215
354.00	357.00	0.39	3,000.0	800	10	1,850.0	15	215

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
13 Jts 5 7/8" HWDP	5 7/8	4.250	120.96	259.51
X/O	7	2.810	0.79	138.55
2 Jts 8 1/4" DC	8 1/4	2.810	18.75	137.76
8" Drilling Cougar Jar	8	3.500	5.84	119.01
5 Jts 8 1/4" DC	8 1/4	2.810	47.24	113.17
X/O	9 5/8	3.000	1.22	65.93
3 Jts 9 1/2" DC	9 1/2	3.000	28.23	64.71
9 1/2" NMDC	9 1/2	3.100	9.44	36.48
9 1/2" UBHO Sub	9 1/2	3.000	0.83	27.04
9 1/2" Float sub (F.valve inside)	9 1/2		0.77	26.21
9 1/2" Pulser Sub	9 1/2		2.15	25.44
9 1/2" NMDC	9 1/2	4.000	9.46	23.29
9 1/2" Restrictor Sub with restrictor	9 1/2	4.250	0.92	13.83
9 1/2" PNMDC	9 1/2		3.05	12.91
11 1/4" Motor (AKO: 1.83° , Sleeve: 23 1/4")	11 1/4		9.25	9.86
24" Mill Tooth Bit	24		0.61	0.61

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
190.86	210.30	0.70	190.86	-1.01	-0.59	1.12	0.11	0.11
227.45	219.40	3.40	227.42	-2.04	-1.39	2.42	2.22	2.21
256.00	226.20	6.10	255.87	-3.74	-3.02	4.78	2.89	2.84
285.40	226.60	8.90	285.01	-6.39	-5.80	8.61	2.86	2.86
314.36	227.00	10.00	313.58	-9.64	-9.27	13.37	1.14	1.14
343.44	228.00	10.90	342.18	-13.20	-13.16	18.64	0.95	0.93
372.64	227.50	12.50	370.77	-17.19	-17.54	24.56	1.65	1.64

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General Remarks

Comment

>>> Requirements: (23" Stabilizer) + (10 1/8" Shock Sub).

>>> Sent:
*** Yaghoub to Kish: (COR 8" Drilling Jar) + (SW, DW, CMT & Bentomite Sample) + (Garbage).

>>> Received:
*** Navis King f/ Kish: (40 MT Barite) + (62 jt of 18 5/8" casing + 1 x Float Shoe & Float Collar) + (COR Fishinf Tools: 12 1/4", 14" & 16" Junk Mill + 15" & 11" RCJB + 7 3/4" Fishng Jar,Type Z + 7 3/4" Accelerator Jar + 7 3/4" Bumper sub + 6 3/8" Taper Tap + 8 1/4" Casing Spear + Spear Stop Sub for 13 3/8" Casing + 15" Stop ring + 15" Impression Block + 8" Safety Joint +15" Fishing Magnet + Marine Spear for 13 3/8" to 20" Casing + Spaer Grapple w/ integral stop ring for 13 3/8" Casing + 11 3/4" Over Shot + 11 3/4" x 36' & 60" Over Shot Extension + 15" Lip Guide + 10", 9 7/8", 9 5/8", 9 1/2" & 9 3/4" Spiral grapple + 9 3/8", 9 1/4", 8 1/8", 8 1/4", 8", 7 7/8", 7 3/4", 7 3/8", 7 1/8", 7", 6 5/8", 6 1/2", 6 1/4", 6", 5 7/8", 5 3/4", 5", 5 1/4" & 6 3/8" Basket Grapple).

*** Borkan f/ Kish: (HPOGC: 11 1/4" Motor + Hydraulic Oil).